Notes on *Cascading Behavior in Networks*

1 Motivation

Reasons behind studying cascading behavior include:

1. Formalize how new ideas spread using quantitative models
2. Help us predict the success/failure of new innovations in their early stages
3. Shape the underlying process of the model to increase/reduce changes of success

Question. How can a behavior adopted by a small group of people spread through an entire population, and how can we mathematically predict or optimze this spread?

1.1 Social Network

Definition. A *social network* is a graph $G = (V, E)$, where:

- $v \in V$ is an individual in the population
- $(v, w) \in E \Leftrightarrow v$ is friends with w.

2 Game Theory: Networked Coordination Games

2.1 Basic Setup

Each node chooses between A, the *old* behavior, and B, the *new* behavior.

Let $0 < q < 1$ be the payoff two nodes receive if they both choose the old behavior A. For each $(v, w) \in E$:

- Both choose $A \Rightarrow$ Reward of q
- Both choose $B \Rightarrow$ Reward of $1 - q$
- Mismatch $\Rightarrow$ No reward

Strategy. A node v should adopt B if at least a q fraction of its neighbors have done so, i.e. $d_v^B > q \cdot d_v$.

Intuition. A smaller q makes a new behavior *easier* to spread, because fewer neighbors need to adopt it before switching becomes worthwhile.

2.2 Modeling with Time

Definition. Each node updates its choice of A or B in response to the behaviors of its neighbors during each time step $t = 1, 2, 3, \ldots$ (either from $A \to B$ or $B \to A$). We define the following sets:

- $S =$ Nodes initially adopting B
- $h_q(S) =$ Nodes adopting B after one round with threshold q
- $h_q^k(S) =$ Nodes adopting B after k rounds

Question. Starting with S, how many time steps does it take to convert (almost) all of the nodes to B?

2.3 Contagion

Definition. A set S is *contagious* if the new behavior originating at S eventually spreads to all nodes.

- It is easier for S to be contagious when q is small.

Definition. The *contagion threshold* of G is the maximum q for which there is a finite contagious set.

2.4 Progressive VS. Nonprogressive Contagion

Nonprogressive Contagion. Nodes can switch $A \to B$ or $B \to A$.

Progressive Contagion. Nodes can only switch $A \to B$.

- $\bar{h}_q(S)$ = Nodes adopting B after one round
- $\bar{h}_q^k(S)$ = Nodes adopting B after k rounds

Theorem. For any graph G, $\exists$ a finite contagious set w.r.t h_q iff $\exists$ one w.r.t. $\bar{h}_q$.

- i.e. both progressive and nonprogressive contagion models have the same contagion threshold.

Theorem. The contagion threshold of any graph is at most $\frac{1}{2}$.

- Intuition: If $q > \frac{1}{2}$, then there will never be enough "incentive" to change to the new behavior

3 More General Models

3.1 Linear Threshold Model

Definition.

- On each edge (v, w), the edge weight $b_{vw} \geq 0$ = the influence that v exerts on w
- Each node v chooses a threshold $0 \leq \theta_v \leq 1$ uniformly at random, where θ_v = the weighted fraction of v's neighbors that must adopt the behavior before v does

Given time steps $t = 1, 2, 3, \ldots$, any inactive node v becomes active if its fraction of active neighbors (denoted as $N(v)$) exceeds its threshold, i.e.

$$\sum_{\text{active } w \in N(v)} b_{vw} \geq \theta_v.$$

3.2 General Threshold Model

Definition. A generalization of the Linear Threshold Model.

- Each node v has a monotone function $g_v(\cdot)$ defined on subsets of $N(v)$, where for any $X \subseteq N(v)$, $0 \leq g_v(X) \leq 1$ is defined.

Each inactive node v becomes active if its set of currently active neighbors satisfies $g_v(X) \geq \theta_v$.

Cascade Model. A probabilistic version of the General Threshold Model (they are equivalent, just syntactically different).

4 Influential Set of Nodes

Question. If we can choose the initial adopters of a new behavior, which people should we select to maximize the eventual spread?

Definition. For a set S of initially active nodes, the *influence function* $f(S)$ is the expected number of active nodes at the end of the process.

- The question of "How to maximize $f(S)$ over all sets S of size k?" is NP-hard.

Evaluating

$$f(s) = \mathbb{E}[\text{eventual } \text{ of active nodes}]$$

may also be computationally intractable. Instead, we can estimate this through random simulations.

Definition. For R simulations, let $A_r(S)$ be the number of nodes eventually activated in simulation r. Then,

$$\hat{f}(S) = \frac{1}{R}\sum_{r=1}^{R} A_r(S)$$

is a *Monte Carlo estimate* of the expected influence.

4.1 Submodularity

Definition. A function $f : 2^V \to \mathbb{R}$ is *submodular* if adding an element to a set Y causes a smaller marginal improvement than adding the same element to a subset of Y. Formally,

$$\forall X \subseteq Y. v \notin Y. f(X \cup \{v\}) - f(X) \geq f(Y \subseteq \{v\}) - f(Y).$$

- $f(X \cup \{v\}) - f(X)$: the marginal benefit of adding v to set X.
- Submodularity says that the benefit of adding x decreases as the starting set becomes larger (i.e. adding adding one more influencer helps more when we start with fewer influencers).

Theorem. (A greedy approximation)
Suppose f is nonnegative, monotone, and submodular. Let S^* be an optimal set of k nodes that maximize f. We consider this greedy algorithm:

1. Start with $S_0 = \emptyset$.
2. At iteration i, choose
$$v_i \in \arg\max_{v \notin S_i} \{f(S_i \cup \{v\}) - f(S_i)\},$$
i.e. the node with the largest marginal increase in f, and update $S_{i+1} = S_i \cup \{v_i\}$.

Let S be the final result of this greedy algorithm. Then,

$$f(S) \geq (1 - \frac{1}{e}) \cdot f(S^*) \geq 0.63 \cdot f(S^*).$$

Theorem. (Local diminishing returns and global submodularity)
If every incremental activation function p_v in the Cascade Model exhibits diminishing returns, then the global influence function f is submodular.

- Intuition: If each individual node experience influence in a way that exhibits diminishing returns, then the entire network experiences influence in the same way.

Theorem. (Special case for the Independent Cascade Model)
For any instance of the Independent Cascade Model, the resulting influence function f is submodular.

Theorem. (General Threshold Model and submodularity)
If every threshold function g_v in the General Threshold Model is submodular, then the resulting global influence function f is submodular.

- This generalizes the theorem regarding Cascade Models and submodularity.